MACHINE LEARNING ALGORITHMS · ENDTERM

Predicting Apartment Asking Prices in Astana

Log-price regression, K-Means, PCA and a neural network on real Krisha.kz listings

≈ 0.88

R^2 of the best model, Gradient Boosting

≈ 9%

median error, against 34% for the baseline

4

market segments found by K-Means

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01 · PROBLEM

Can we predict an Astana asking price?

Predict the asking price of an apartment from its area, location, rooms, floor, building and amenities.

01

Task

Supervised regression on tabular listing data

02

Target

price in KZT; models learn log1p(price) because prices are right-skewed

03

Success metrics

Log RMSE, MAE and R^2; KZT MAE and median absolute percentage error (MedAPE)

A decision-support estimate, not an official valuation · test set used once

02 · DATASET AND PREPARATION

Real Krisha.kz listings, cleaned without leakage

≈ 3.9k
listings after cleaning

15
numeric features, plus categorical

Kaggle
turarr, scraped from Krisha.kz

Cleaning

- Parse numbers; split floor pairs like “5 of 9”
- Drop IDs, free text and price-like leakage fields
- Remove duplicates and columns over 60% missing
- Range checks on area, rooms, year and floors
- Trim the outer 1% of prices
- Impute only inside the pipeline, on train data

Engineered features

`house_age` `is_first_floor` `is_last_floor`
`floor_ratio` `area_per_room` `distance_to_centre_km`

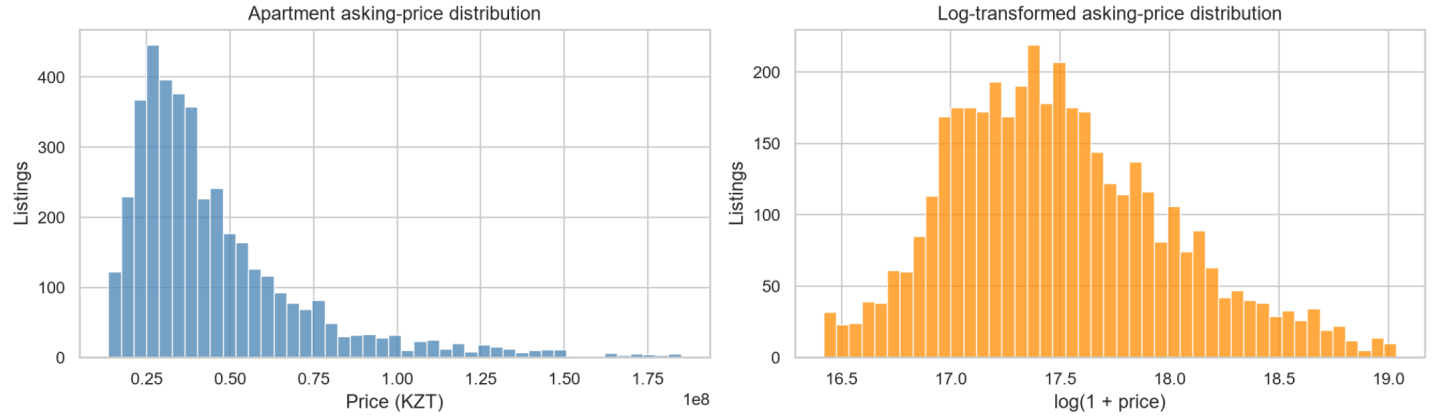
Built from listing attributes only: no feature uses the target.

Limits: asking prices, duplicates, seller-entered fields, one collection period.

kaggle.com/datasets/turarr/astana-real-estate-dataset · CSV loaded with pandas

03 · EXPLORATORY ANALYSIS

Prices are right-skewed, so models learn log-price



14M–185M

KZT price range kept after trimming

25–30M

KZT: the most common asking price

Long upper tail

Most listings are below 75M KZT; a few luxury flats reach 150M and above.

log1p fixes the shape

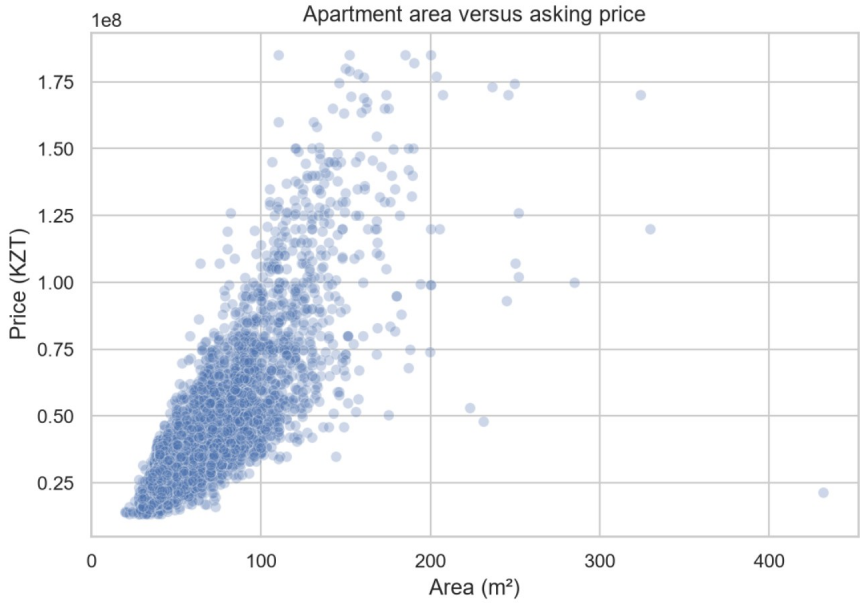
The log scale is close to symmetric, which suits regression and keeps luxury listings from dominating.

Evaluate in KZT too

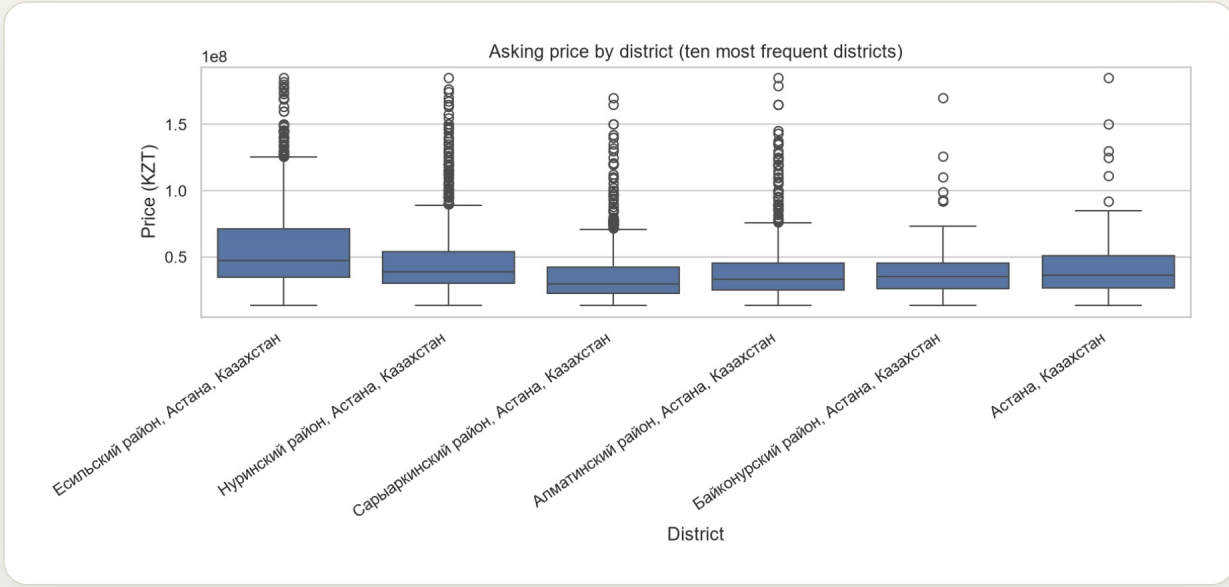
Predictions are converted back, so MAE and MedAPE are reported on the real price scale.

Outer 1% of prices trimmed; the log distribution still has a mild right tail

Area and district drive asking prices



Area is the main driver: correlation ≈ 0.8 with price, and the spread widens above 100 m².



District matters: Esil has the highest median (≈ 47M KZT) and the widest range; Saryarkin is the cheapest (≈ 30M).

05 · EVALUATION DESIGN

70 / 15 / 15 split with no leakage

Train 70%
fit preprocessing, tune with 5-fold CV

Val 15%
pick model

Test 15%
used once

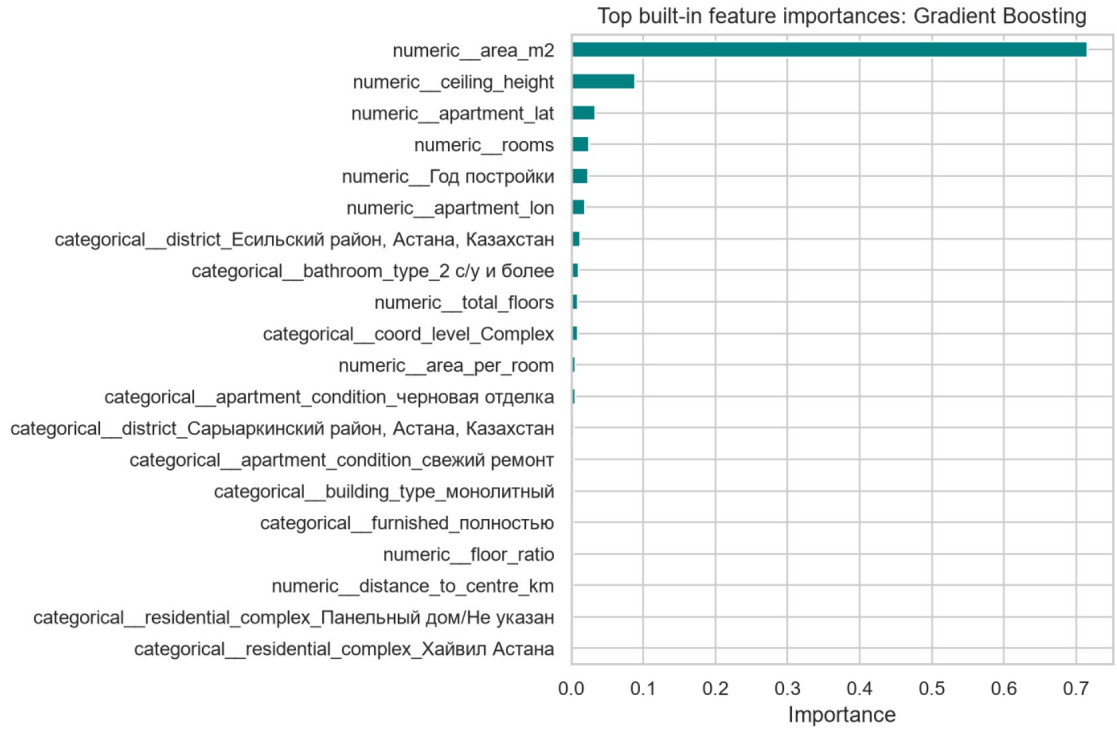
1
Stratified by price
Ten quantile bins of log-price keep cheap, middle and expensive listings in every split.

2
Pipeline-only fitting
Imputer, scaler and one-hot encoder are fitted on training data, and on training folds inside CV.

3
One clean final test
The chosen model is refit on train plus validation and scored on test a single time.

Random seed 42 everywhere · rare categories grouped (minimum frequency 10)

Midterm baselines, tuned and extended

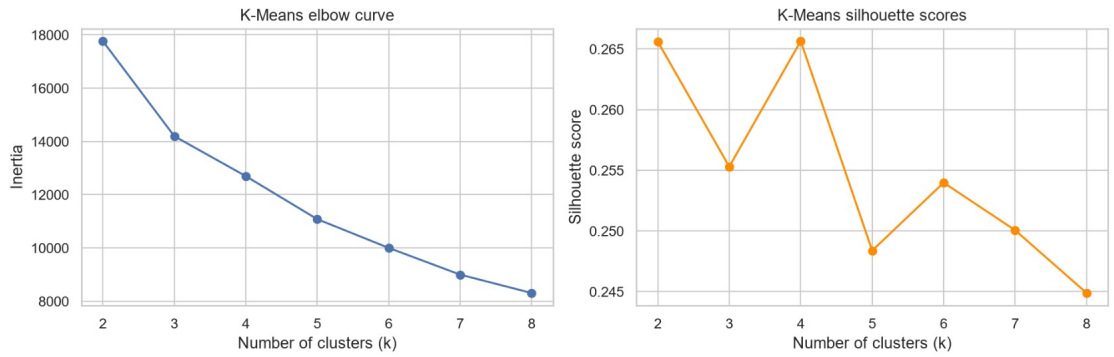


Model	Tuning (5-fold CV, train)
Median baseline	None: reference
Linear Regression	None: benchmark
Decision Tree	Grid: depth, leaf
KNN	Grid: k, weights
Random Forest	Random search ×12
Gradient Boosting	Random search ×12

area_m2 carries ≈ 71% of Gradient Boosting importance; ceiling height ≈ 9%.

First four models are the midterm set · importance describes predictive use, not causes

K-Means finds four market segments



Setup and choice of k

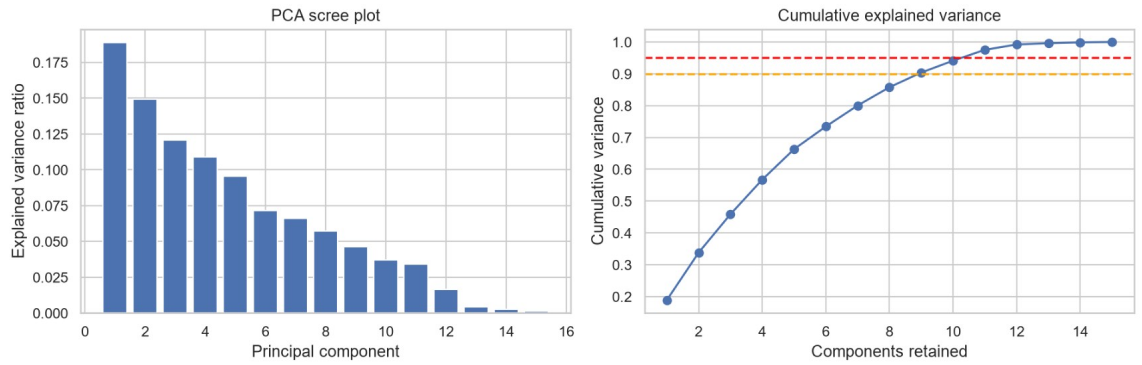
- Features: price, area, rooms, house age, floor ratio, coordinates
- Median imputation, then standardization (distance-based method)
- k = 2–8, k-means++ start, 20 restarts
- Best silhouette ≈ 0.27 at k = 4 (k = 2 is tied); the elbow has no sharp kink, so k = 4 gives richer segments



Cluster 2: large, expensive flats (about 100–250 m²). **Clusters 0, 1, 3** overlap in size and price and differ in age, floor and location.

Exploratory only: clusters are not used as inputs to the regression models

PCA: variance is spread across many components

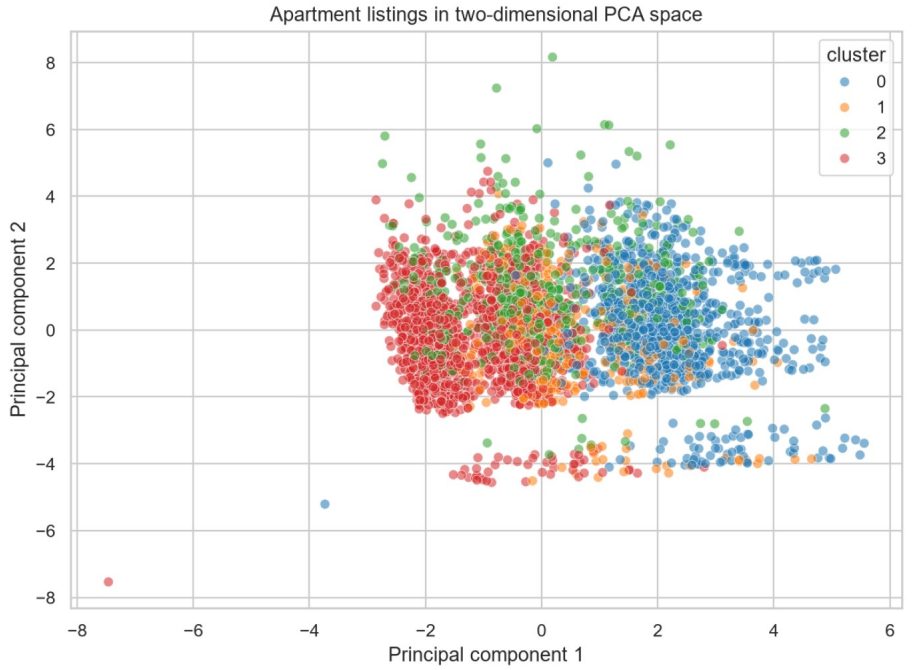


≈ 34%
variance in 2 components

9
components for 90%

11
components for 95%

Fitted on 15 standardized, median-imputed numeric training features. No dominant axis: the features carry mostly independent information, so PCA compresses little.



2D view: cluster 3 sits left and cluster 0 right along PC1, with heavy overlap, matching the weak silhouette of ≈ 0.27.

09 · NEURAL NETWORK

Feedforward ANN: what training showed

Input
preprocessed

→

Dense 128
ReLU · drop 0.2

→

Dense 64
ReLU · drop 0.2

→

Dense 32
ReLU

→

Output 1
linear



Setup

MSE loss, Adam at 0.001, batch 32, early stopping and learning-rate halving on plateau

Convergence problem

Loss starts near 66: outputs begin near 0 while log-prices are near 17.5. Validation bottoms at ≈ 1 around epoch 12 and then rises; training stopped at 33 epochs.

Result and next step

RMSE ≈ 1.0, worse than the baseline (≈ 0.50). Likely fix: standardize the target, then retrain.

Tabular data, so an ANN is the right deep-learning model; a CNN or NLP model would not apply

10 · MODEL COMPARISON

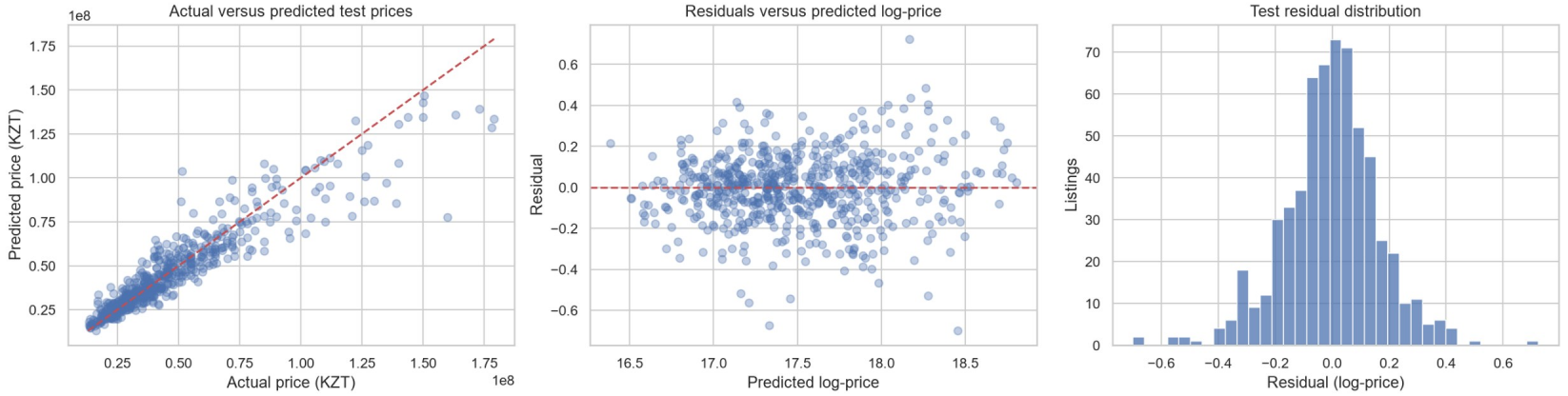
Gradient Boosting wins on every metric

Model, validation set	RMSE (log)	R^2	MAE (KZT)	MedAPE
Gradient Boosting	0.16	0.88	5.7 M	**9.3%**
Random Forest	0.17	0.87	6.0 M	9.4%
Linear Regression	0.21	0.81	7.6 M	11.9%
Decision Tree (tuned)	0.22	0.80	8.1 M	12.8%
KNN (tuned)	0.24	0.76	8.2 M	12.9%
Median baseline	0.50	0.00	17.8 M	34%
Neural network (ANN)	**1.02**	**−3.2**	**29.4 M**	**62%**

Same split and metrics for all models · values read from the validation charts (±0.01)

11 · FINAL TEST AND ERROR ANALYSIS

Where the best model errs on the test set



Unbiased on average

Residuals centre on 0; most fall within ±0.2 log-price (about ±20%).

Premium flats underpredicted

Above ≈ 100M KZT, predictions fall below the diagonal.

Even spread

Error does not grow with predicted price; a few outliers reach ±0.7.

Final test metrics: RMSE (log) **0.167** · R^2 **0.89** · MAE **0.125** KZT · MedAPE **9.27%**

Best model refit on train plus validation, scored once on the test set

12 · LIMITATIONS AND ETHICS

What the model cannot tell you

Data limitations

- Asking prices differ from negotiated sale prices
- Duplicates, missing and seller-entered errors
- One snapshot in time, so no time-aware validation
- Renovation, view and building details are missing, which hurts premium flats

Responsible use

- A decision-support estimate, not an official appraisal
- Feature importance shows predictive use, not causes
- Accuracy varies by price segment and district
- A prediction is not a fair or guaranteed price

Seen in our own results: premium listings are underpredicted, and the unscaled-target ANN failed. Next: verified sales, quality data, time-aware validation.

Ethical angle: an automated price anchor can mislead buyers and sellers if over-trusted

13 · CONCLUSIONS

Four findings from the evidence

$R^2 \approx 0.88$

Gradient Boosting is the best model

Median error ≈ 9% against 34% for the baseline; Random Forest is within 0.01 RMSE.

≈ 71%

Area dominates the price

Then ceiling height ≈ 9%, latitude, rooms and build year; importance is not causality.

4 clusters

Weak market structure

Silhouette ≈ 0.27, one clear large-flat segment; PCA needs 9 components for 90% variance.

R^2 −3.2

The ANN failed, for a fixable reason

The log-price target was not standardized; next step is to standardize it and retrain.

Answer: yes, asking prices are predictable to about 9% median error, least accurately for premium listings.

Validation results; final test metrics on slide 12

Who did what

Nurkhan Kuanyshov

- Data pipeline: cleaning, feature engineering, EDA
- Leakage-safe split and preprocessing pipeline
- Midterm baselines, Decision Tree and KNN tuning, K-Means

Presents slides: 1–8

Bakytzhan Jangazy

- Random Forest and Gradient Boosting tuning, PCA
- Neural network: architecture, training, diagnosis
- Final comparison, test evaluation, error analysis, README

Presents slides: 9–15

Thank you. Questions?

Code, README and the contribution record are in the project repository